Thermodynamics Example Test



Written by indian

Team Name: _____

Student Name(s): _____

Please do not flip this test over until instructed to.

Section 1: Conversions and Formulas

Round to the nearest hundredth if needed.

1. Convert 56 Fahrenheit to Celsius.
2. Convert 108 Celsius to Kelvin.
3. Convert 98 Rankine to Celsius.

4. Convert 5 Calories to Joules.
5. Convert 8 large calories to small calories.

Section 2: Heat

1. Describe what the triple point is in the context of states of matter.

2. Match the following to one of these terms: temperature, thermal energy, heat
    a. The transfer of thermal energy between substances due to a temperature difference.
    b. The average energy of individual particles within a substance
    c. The total energy of all particles in a substance combined

3. Match the following to one of these terms: entropy, enthalpy, spontaneity
    a. The measure of disorder in a system
    b. How easy it is for something to happen without external input
    c. The sum of a system's internal energy plus the product of its pressure and volume.

4. Select all statements that are true for this reaction: exothermic, with an increase in entropy
    a. ΔG is always negative
    b. ΔG is always positive
    c. System is spontaneous at all temperatures
    d. System is not spontaneous at all temperatures
    e. System is spontaneous at low temperatures
    f. System is spontaneous at high temperatures

5. Select all statements that are true for this reaction: exothermic, with a decrease in entropy
    a. ΔG is always negative
    b. ΔG is always positive
    c. System is spontaneous at all temperatures
    d. System is not spontaneous at all temperatures
    e. System is spontaneous at low temperatures
    f. System is spontaneous at high temperatures

6. Select all statements that are true for this reaction: endothermic, with an increase in entropy
    a. ΔG is always negative
    b. ΔG is always positive
    c. System is spontaneous at all temperatures
    d. System is not spontaneous at all temperatures
    e. System is spontaneous at low temperatures
    f. System is spontaneous at high temperatures

7. Select all statements that are true for this reaction: endothermic, with a decrease in entropy
    a. ΔG is always negative
    b. ΔG is always positive
    c. System is spontaneous at all temperatures
    d. System is not spontaneous at all temperatures
    e. System is spontaneous at low temperatures
    f. System is spontaneous at high temperatures

8. Provide an example of conduction.

9. Provide an example of convection.

10. Explain why hot air rises in terms of density and molecular motion.

11. Provide an example of radiation.

12. Is feeling warmth from sunlight outside primarily an example of radiation? Explain.

Section 3: Laws, laws, laws, and more laws

1. In the First Law of Thermodynamics, there are two versions of the equation. Please write both of the versions, and write the condition which needs to be met to use that specific version.

2. What does ΔU stand for in this equation?

3. Who helped formulate the First Law of Thermodynamics?

4. Fill in the blanks: The Second Law of Thermodynamics states that for any spontaneous process, _____ _____s over time.
    a. Enthalpy, increases
    b. Enthalpy, decreases
    c. Entropy, increases
    d. Entropy, decreases

5. What is the formula for the Second Law of Thermodynamics (hint: integrals… scary)

6. What does the Clausius Statement say?

7. What does the Kelvin-Planck Statement say?

8. What does the Third Law of Thermodynamics State?

9. What unit must the temperature be in?
    a. Fahrenheit
    b. Celsius
    c. Kelvin
    d. Rankine

10. Wait… is there a *zeroeth* law?
    a. True
    b. False

11. If there is one, what does the zeroeth law state?

12. When solving a question using the Ideal Gas Law, in what SI base unit must V be in?

a. Grams
b. Cm^3
c. M^3
d. Liters

13. What are the two conditions that must be met in order for a gas to be “ideal”?

14. Derive the ideal gas law.
    a. Solve for molar mass.

    b. Solve for density.

Section 4: Systems and Processes

1. A pot of boiling water is on a stove, without a lid. Identify what type of thermodynamic system this is.
    a. Open System
    b. Closed System
    c. Isolated System
    d. Diathermic System
    e. Adiabatic System

2. A perfectly sealed vacuum thermos has hot chocolate inside of it. Identify what type of thermodynamic system this is.
    a. Open System
    b. Closed System
    c. Isolated System
    d. Diathermic System
    e. Adiabatic System

3. A lab flask with a stopper is used for chemical reactions. Identify what type of thermodynamic system this is.
    a. Open System
    b. Closed System
    c. Isolated System
    d. Diathermic System
    e. Adiabatic System

4. A man is pumping a bicycle pump. Identify what type of thermodynamic system this is.
    a. Open System
    b. Closed System
    c. Isolated System
    d. Diathermic System
    e. Adiabatic System

5. There is a sealed water bottle on the table. Identify what type of thermodynamic system this is.
    a. Open System
    b. Closed System
    c. Isolated System
    d. Diathermic System
    e. Adiabatic System

6. Give an example for each of the following processes.
    a. Isobaric Process
    b. Isochoric Process

c. Isothermal Process
d. Adiabatic Process
e. Isentropic Process

Section 5: The Math

Formulas: conduction, convection, and radiation (in that order)

$$\dot{Q} = \frac{kA(T_h - T_c)}{L} \qquad \dot{Q} = hA(T_s - T_\infty) \qquad P = \sigma e A T^4$$

1. Look at the formula for conduction.
    a. Explain all of the variables in the formula.

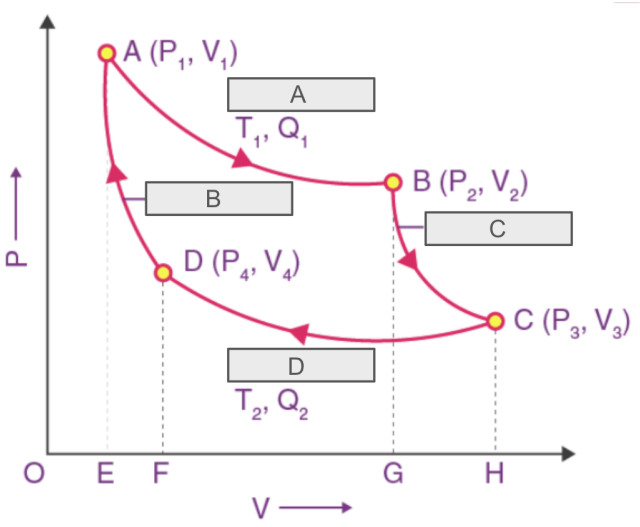
    b. Derive this formula to solve for L.

2. Look at the formula for convection.
    a. Explain all of the variables in the formula.

    b. Derive this formula to solve for A.

3. Look at the formula for radiation.
    a. Explain all of the variables in the formula.

    b. Derive this formula to solve for T.



4. Label this diagram of a Carnot Cycle.
    a. _____
    b. _____
    c. _____
    d. _____

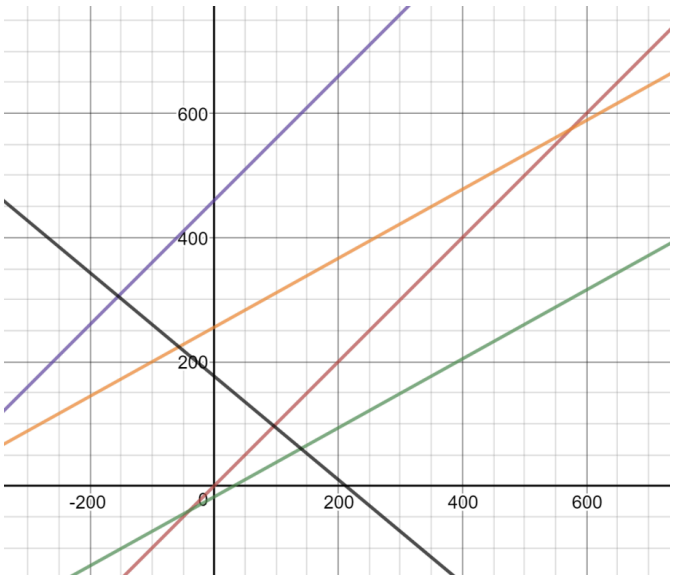
5. What is the thermal efficiency of a heat engine with a heat energy input of 300 Celsius from the hot source, and 200 Celsius of waste energy outputted to the cold sink.

6. List the formulas of the different types of expansion.

a. Linear Expansion

b. Area Expansion

c. Volume Expansion



7. The image above depicts five different temperature scales. Label them.
   a. Black line: _____
   b. Red line: _____
   c. Green line: _____
   d. Yellow line: _____
   e. Purple line: _____

8. A gas occupies 0.020 m^3 at 300 K and 150 kPa. How many moles of gas are present?

9. A wall with thermal conductivity 0.84 W/m·K has an area of 5 m² and thickness 0.20 m. The temperature difference across the wall is 30 K. Find the heat transfer rate.

10. How much energy is required to raise 2.5 kg of water from 20°C to 85°C?

11. A gas initially occupies 2.0 L at 300 K and 1 atm. It is heated to 450 K while pressure increases to 1.5 atm. Find the final volume.

12. Engine A operates between 600 K and 300 K. Engine B operates between 900 K and 450 K. Find the ratio of their efficiency.